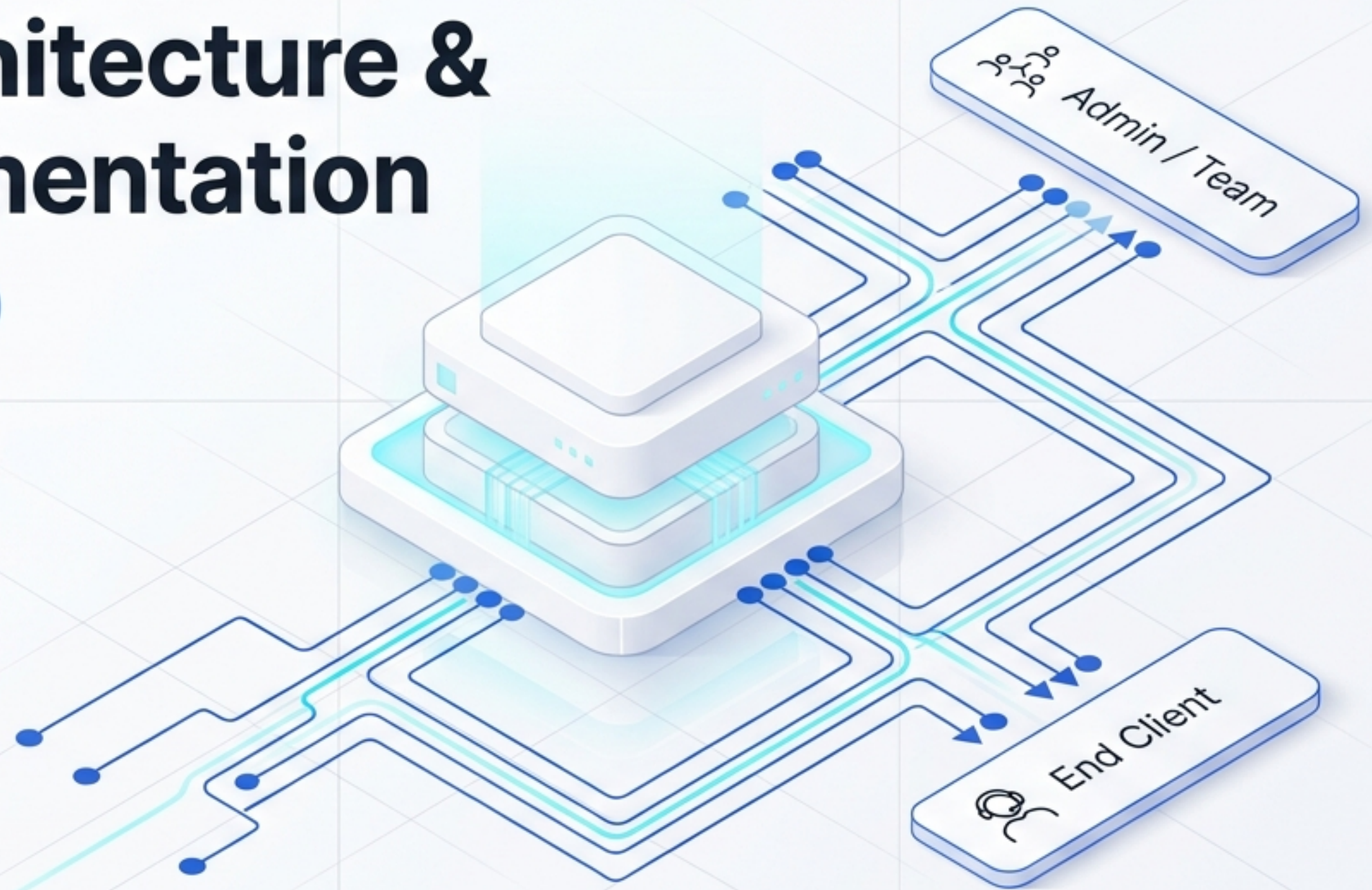


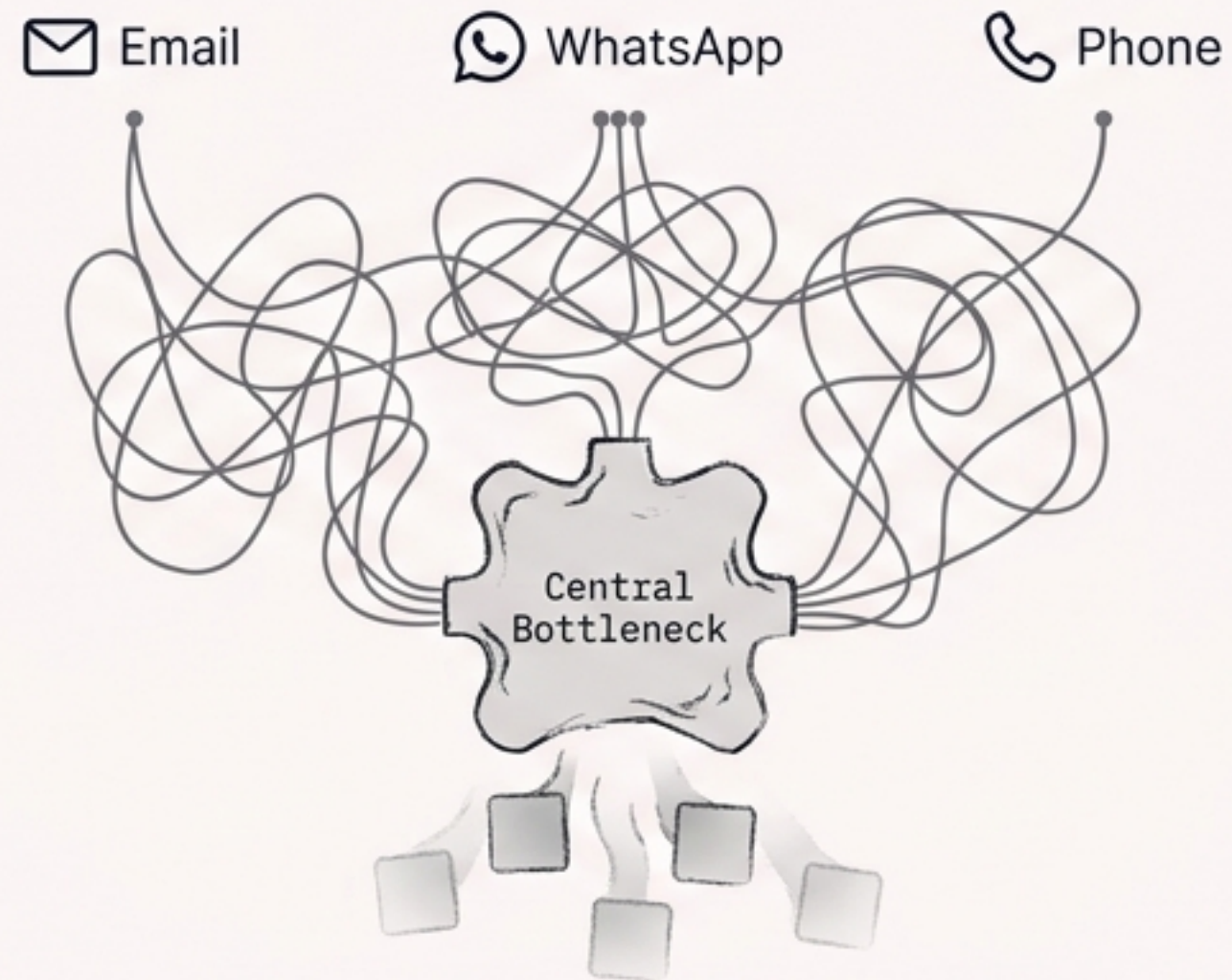
HelpDesk System: Technical Architecture & System Documentation

Platform Blueprint v1.0



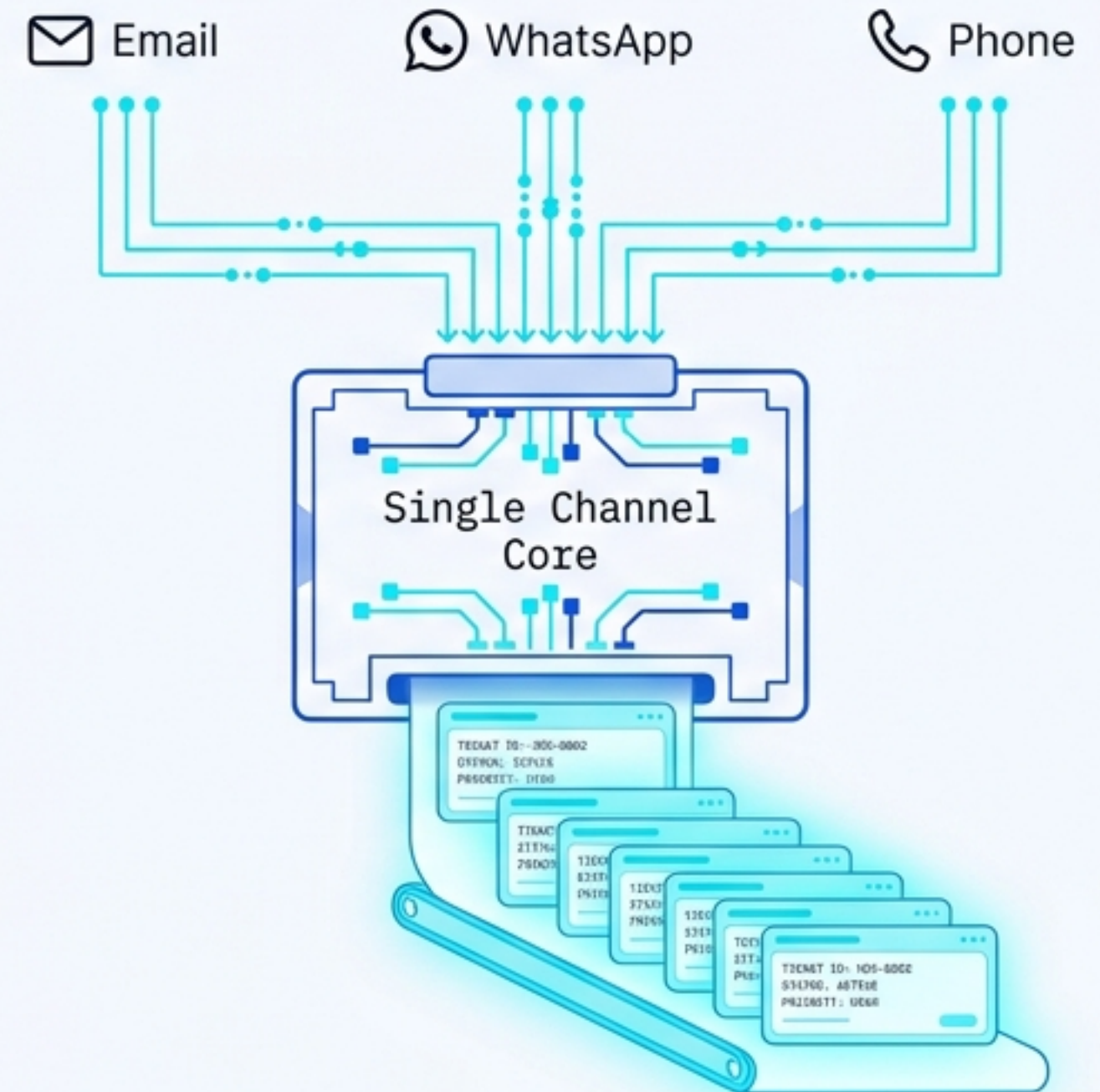
The Shift from Chaos to Controlled Protocol

Decentralized Operations



Impossible to track history, measure resolution time, or guarantee accountability.

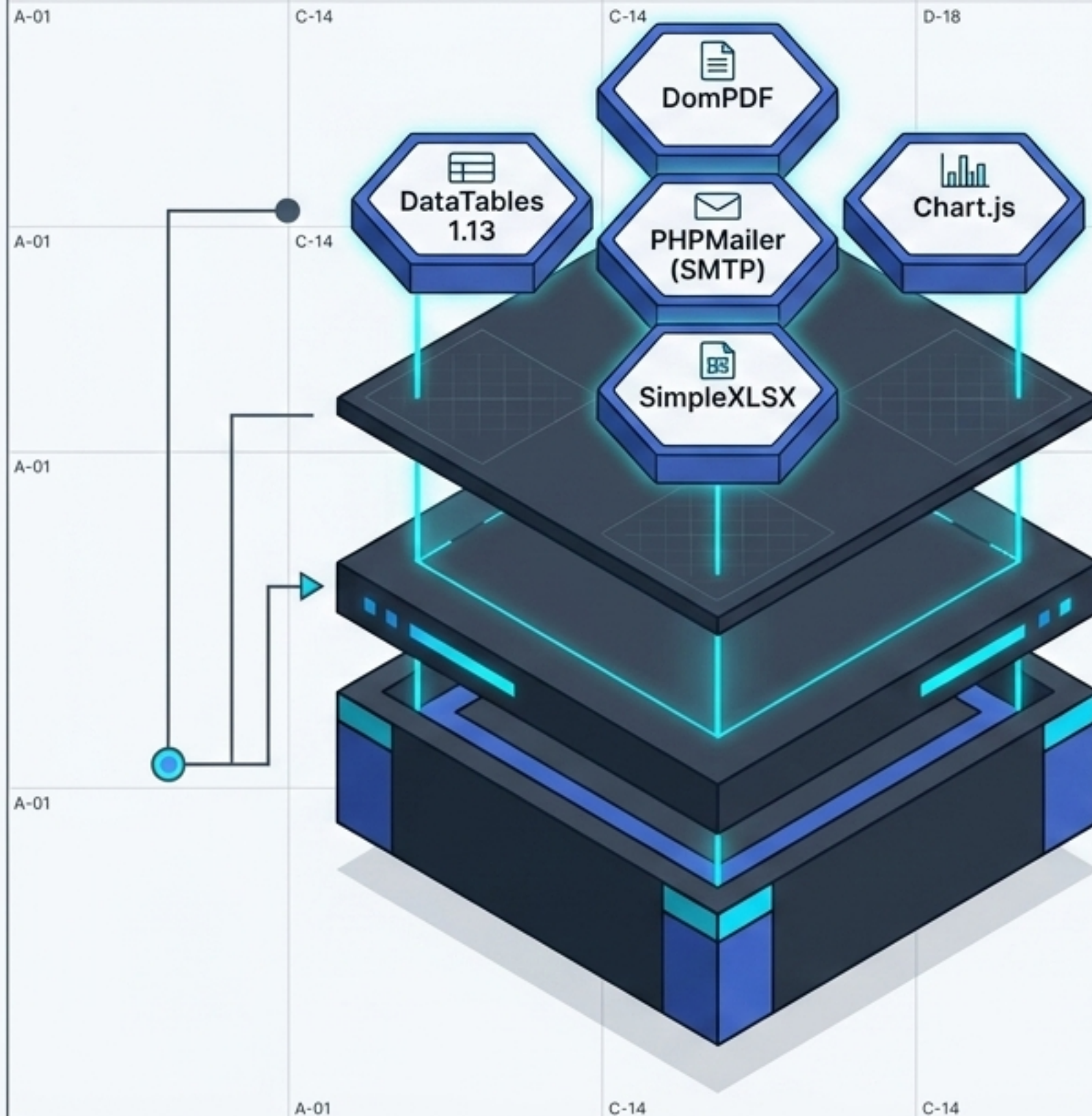
The HelpDesk Solution



Single-channel entry, automated protocol generation, complete operational auditing.

The Technology Stack

A highly structured separation of concerns relying on robust, industry-standard tooling.



Presentation Layer

Logic Layer

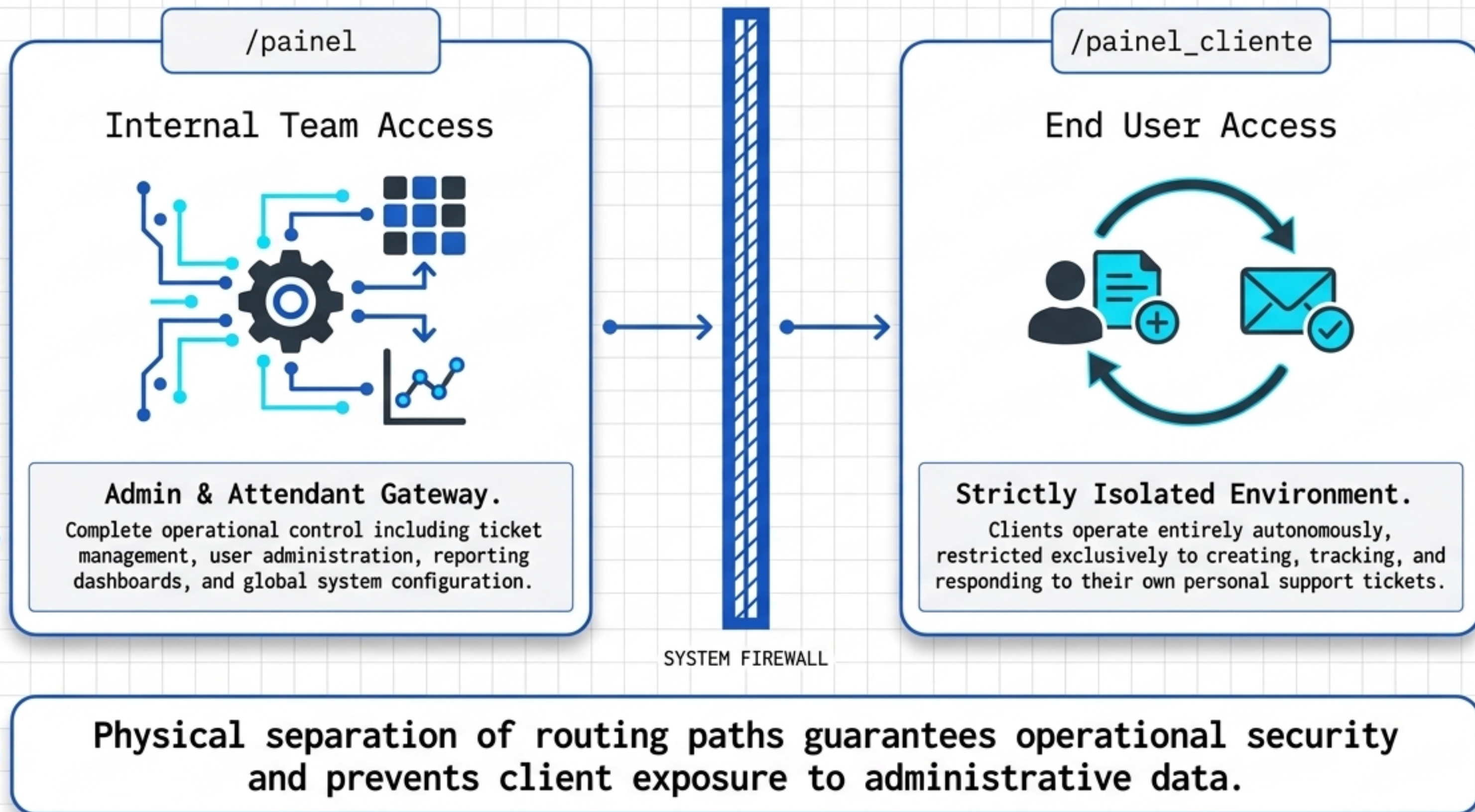
Data Layer

HTML5, CSS3, JS, Bootstrap 5.
Responsive, multi-device frontend.

PHP 8.x.
Simplified MVC architecture and core routing.

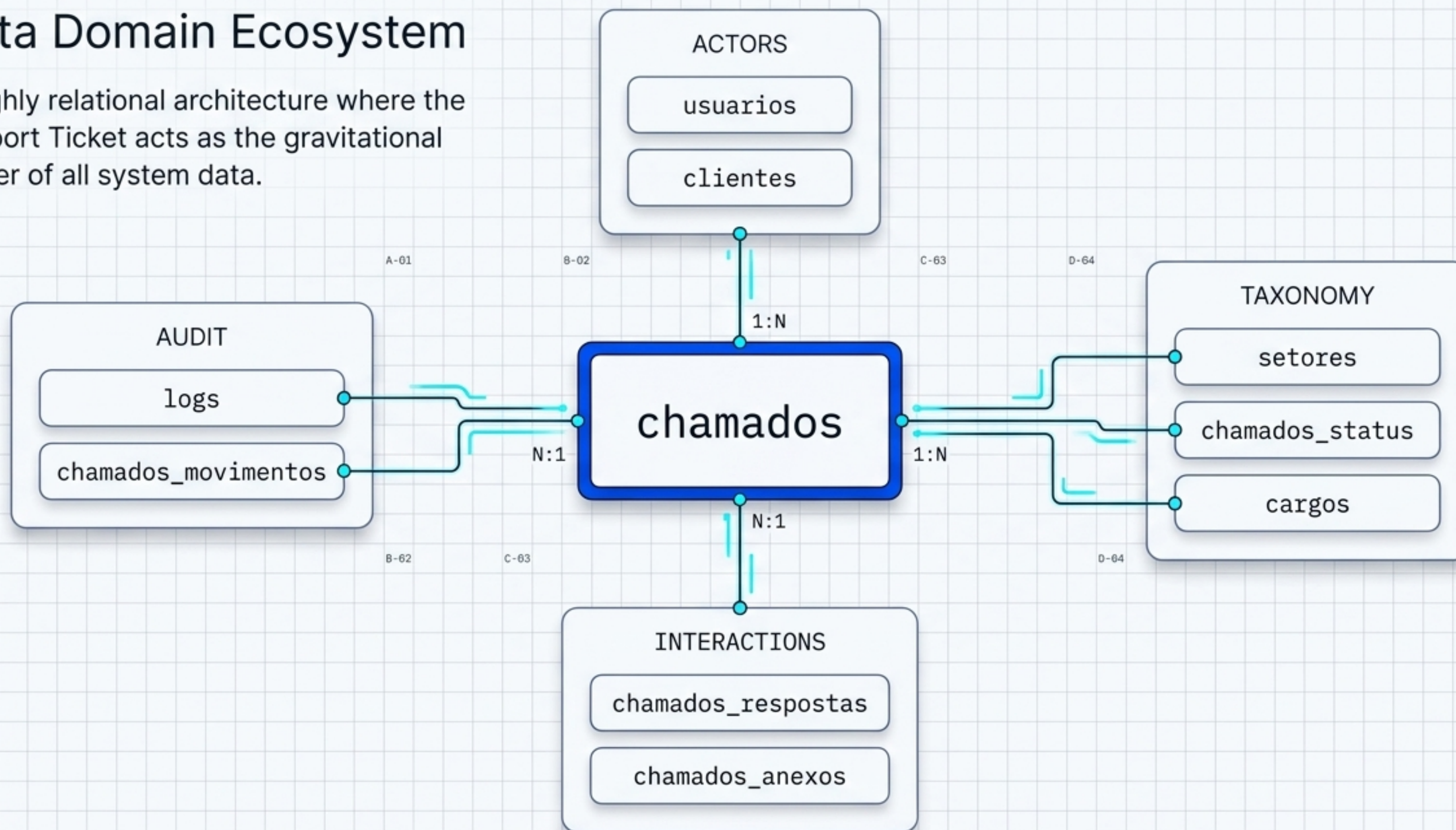
MySQL (InnoDB / utf8mb4).
Fully relational data storage.

The Dual-Panel Paradigm



Data Domain Ecosystem

A highly relational architecture where the Support Ticket acts as the gravitational center of all system data.

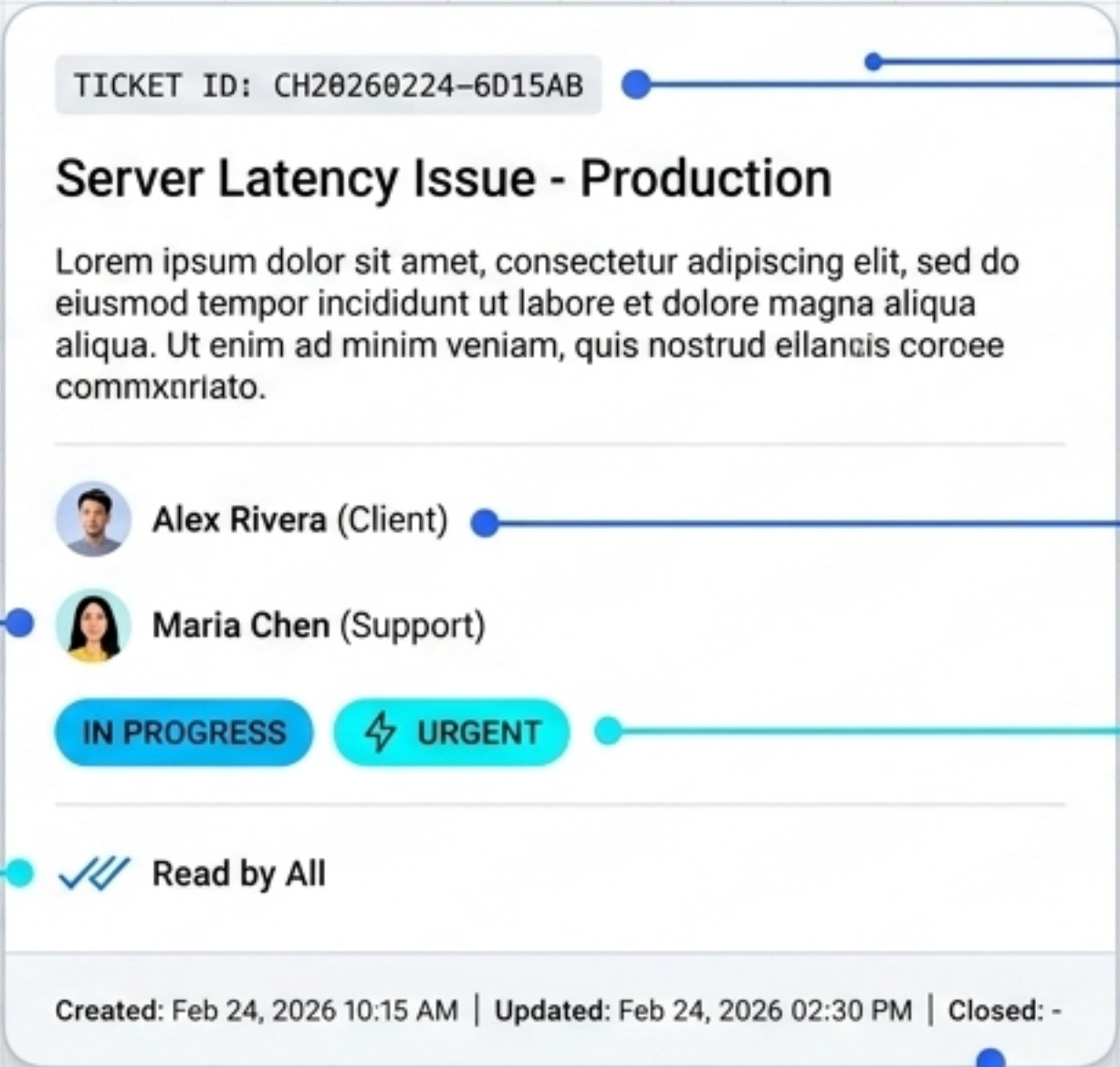


Anatomy of the Core Engine: The Ticket

A-01

A-06

A-01



Foreign Keys :
cliente_id,
usuario_abertura_id,
usuario_responsavel_id,
setor_id

Read Receipts :
visualizado_cliente,
visualizado_suporte

protocolo :
1 Auto-generated unique ID
(e.g., CH20260224-6D15AB)

Foreign Keys :
2 cliente_id,
usuario_abertura_id,
usuario_responsavel_id,
setor_id

State Enums :
3 prioridade (Baixa, Media,
Alta, Urgente) & status_id

Temporal Data :
5 criado_em,
atualizado_em,
fechado_em

B-12

C-63

D-01

Role-Based Access Control

	Admin	Attendant	Basic	Client
View Own Tickets	✓	✓	✓	✓
View Sector Tickets	✓	✓		
Reply to Tickets	✓	✓	✓	✓
Global System Config	✓			
Export Data	✓	✓		
Manage Users	✓			

Enforcement Logic

1. CRUD Validation:
`podeFazer(string $acao)`
Validates granular permissions for 'criar', 'editar', or 'excluir'.

2. Module Visibility:
`podeAcessar(?array $perms, string $menuId)`
Controls dynamic rendering of navigational elements based on user tier.

The Ticket Lifecycle Pipeline

A-01

B-12

C-63

C-03

Client Actions

1. Opening

Web portal entry

Auto-Protocol
Generated
(CHXXXXXXXX)

5. Tracking

Public or
logged-in view

Internal Actions

2. Triage

3. Assignment

4. Support

Internal notes
& replies

6. Closure

PHPMailer SMTP
Notification
Dispatched

Automated System
Movement Logged

A-06

B-12

A-01

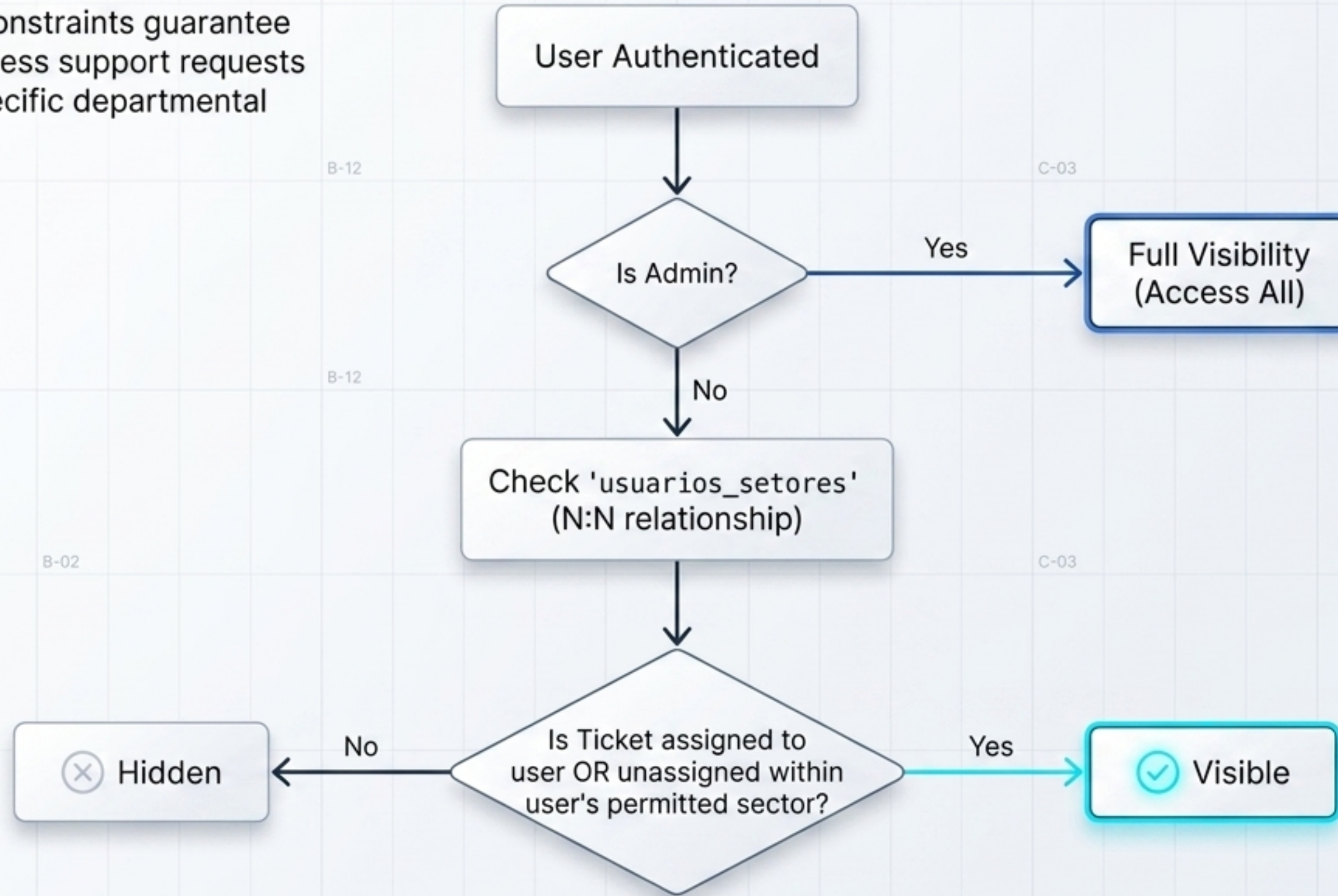
B-02

C-63

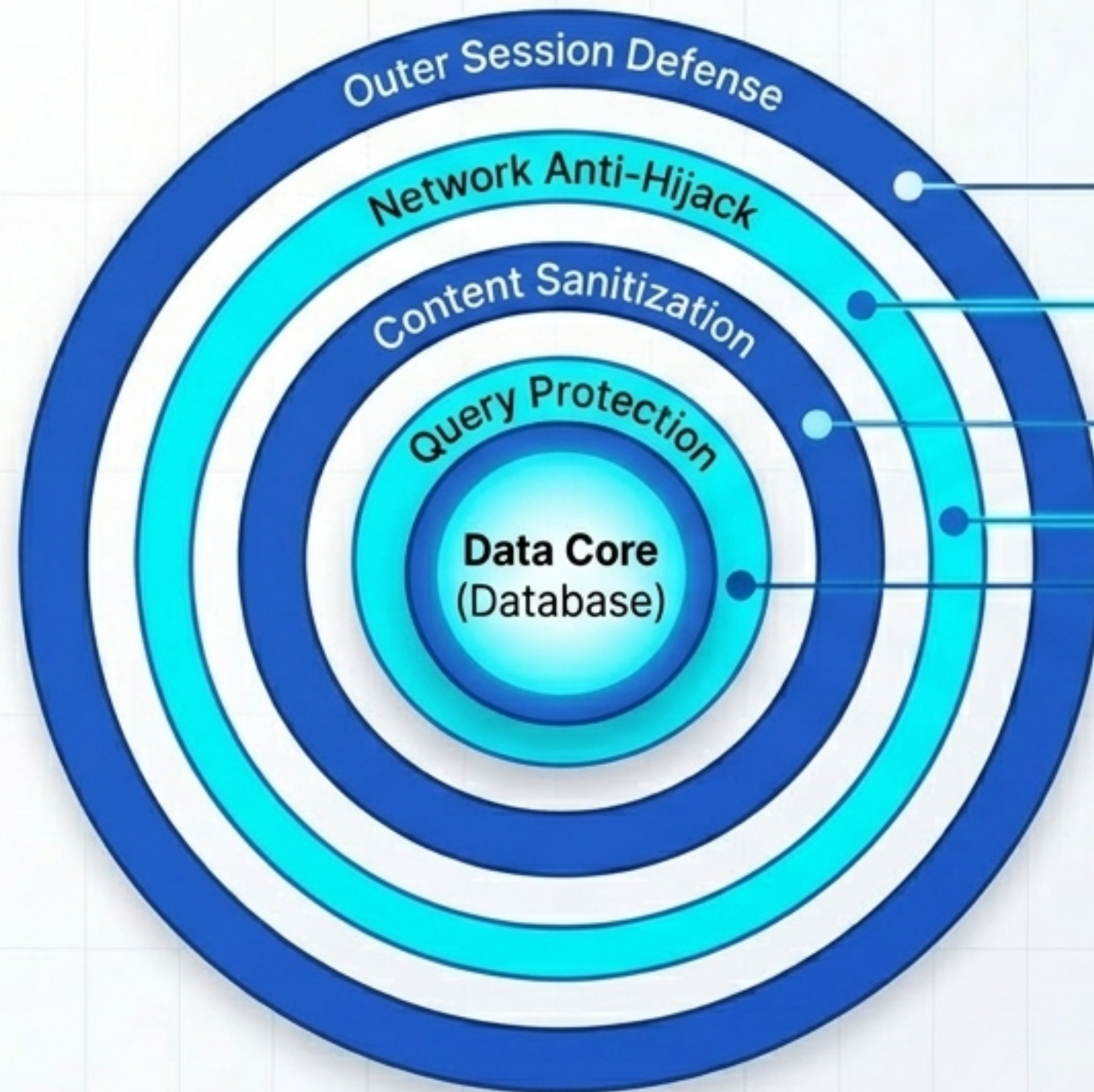
D-01

Sector Routing & Visibility Logic

Dynamic visibility constraints guarantee attendants only access support requests relevant to their specific departmental competencies.



The Security & Defense Framework



1. Outer Session Defense: CSRF Tokens (csrf_token_login) & robust session validation.

2. Network Anti-Hijack: Strict session matching via `$_SESSION['token_login'] == $_SESSION['id']`.

3. Content Sanitization: Strict Output escaping via `esc()` for XSS prevention & 12MB limit
bin2hex renamed safe uploads.

4. Query Protection: Total implementation of PDO Prepared Statements eliminating SQLi vectors.

5. Data Encryption: Irreversible Bcrypt Password Hashing for user credentials.

Total Accountability: Audit & Telemetry

Every human interaction and automated state change is immutably recorded, ensuring zero blind spots in operations.

Table: logs | BIGINT PK

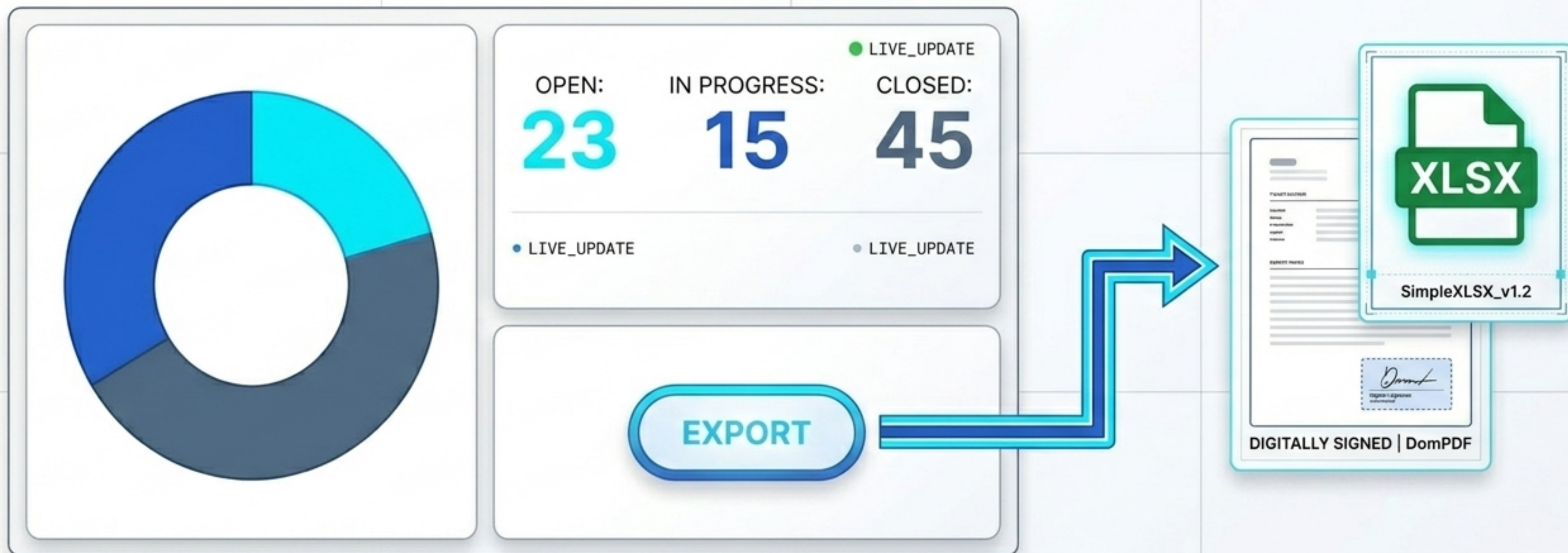
usuario_id	acao	entidade	ip	timestamp
45	login	auth	192.168.1.1	2026-05-12
45	editar	chamados	192.168.1.1	2026-05-12 08:05:44

Table: chamados_movimentos

chamado_id	tipo	status_de	status_para
1024	status	Aberto	Em Análise
1024	atr.	NULL	Atendente B
1024	status	Em Análise	Fechado

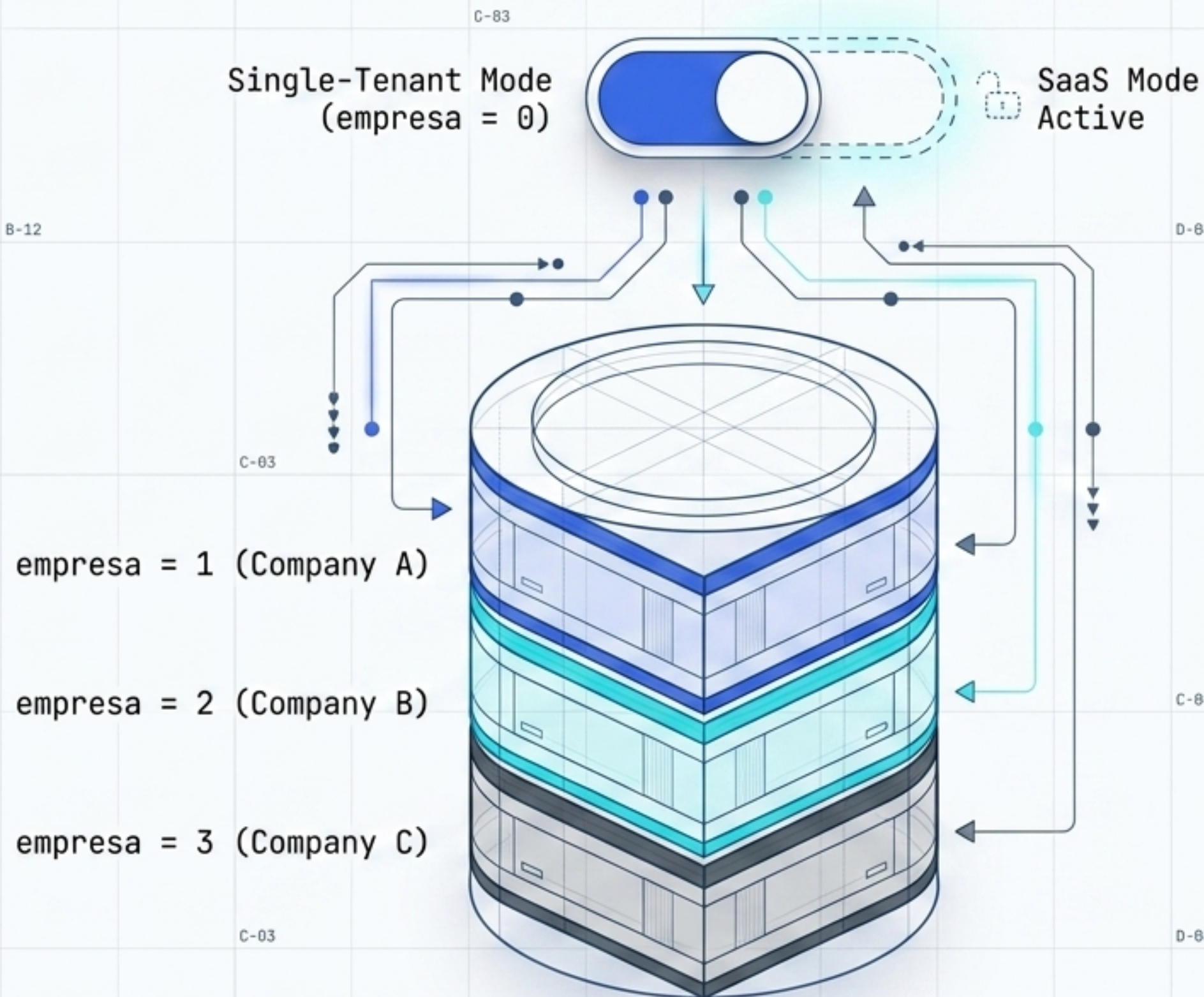
Real-time Analytics & Export

Dynamically calculated metrics alongside enterprise-grade export capabilities, featuring digitally signed PDFs and granular Excel reports.



Future-Proofing: Multi-Tenant Architecture

The foundational schema is pre-configured with tenant identifiers, allowing seamless future transition to a multi-company SaaS model within a single unified deployment.



The Complete Support Ecosystem

A secure, MVC-driven platform that transforms chaotic requests into tracked, actionable, and permanently audited data. Ready for scale.

